



SPIRAL: A Novel Time Series to Image (TS2I) Transformation Method for Vision-Based Anomaly Detection

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Abstract

Time series anomaly detection (TSAD) underpins critical applications in manufacturing, healthcare, finance, and cloud computing. Recently, time series to image (TS2I) transformations have emerged as a promising approach that enables leveraging pretrained vision models for time series analysis. However, the majority of existing TS2I methods were designed for forecasting and classification, and suffer from parameter sensitivity, symmetric redundancy, and quadratic computational complexity. In this study, we introduce SPIRAL, a novel TS2I transformation designed for TSAD. SPIRAL maps time series windows onto a two-arm Archimedean spiral, producing asymmetric representations that preserve temporal locality while eliminating redundancy with linear time complexity with respect to window length. We additionally propose a standardized workflow with ACF-based window selection and point-wise scoring. Extensive evaluation (24,430 experiments) on 23 datasets from the TSB-AD benchmark, comparing SPIRAL against 9 TS2I transformations, 3 vision backbones, and 32 time-domain baselines across 9 metrics, shows that SPIRAL achieves the best average rank among 102 configurations in challenging metrics such as VUS-PR while exhibiting 40% lower training instability than its closest TS2I competitor. Our code is publicly available at: <https://github.com/Smendowski/SPIRAL>.

Keywords Anomaly detection · Time series · Time series to images · Transfer learning · Deep learning · Computer vision

1 Introduction

Time-series anomaly detection (TSAD) is essential for applications such as predictive maintenance, finance, cybersecurity, cloud computing, and healthcare (Zamanzadeh Darban et al., 2024). Yet, reliable detection remains challenging: time series exhibit diverse temporal patterns, frequent distribution shifts, and sensitivity to data representation choices (Zhang et al., 2022; Nawrocki & Smendowski, 2025). These challenges have motivated increasing

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interest in time series to image (TS2I) methods, which transform 1D signals into 2D images (Hatami et al., 2017). Beyond compatibility with computer vision (CV) backbones, TS2I can encode temporal dependencies as spatial patterns, making certain structures easier to capture than in native time-domain form.

While TS2I has shown promise in time series forecasting and classification, its role in TSAD remains insufficiently understood (Ni et al., 2025). Existing TS2I transformations exhibit several limitations when applied to anomaly detection: i) they are general-purpose representations rather than TSAD-oriented mappings; ii) they are parameter-dependent and capture structures that appear in some datasets but not others, making generalization across domains difficult; iii) many exhibit symmetries that introduce substantial redundancy, leading to inefficient use of the image layout; iv) several widely used transformations incur quadratic computational overhead in window length; and v) many cause training instability and poor convergence with pretrained backbones.

In this paper, we propose SPIRAL, a novel model-agnostic TS2I transformation designed specifically for TSAD. SPIRAL is grounded in four core principles: i) parameter-free design requiring no hyperparameter tuning; ii) linear time complexity with respect to window length; iii) asymmetric spatial encoding that eliminates redundancy; and iv) anomaly-centric representation that amplifies deviations for vision-based detection. These properties facilitate stable and effective training of vision backbones while retaining discriminative temporal structure for downstream detectors. Moreover, we introduce a TSAD-oriented preprocessing and evaluation workflow that reduces sensitivity to windowing choices by leveraging autocorrelation function (ACF) analysis to establish an informed window size, and by aligning training and evaluation with point-wise anomaly prediction.

The key contributions of this paper can be summarized as follows:

- C1: We introduce SPIRAL, a novel model-agnostic TS2I transformation specifically designed for TSAD, achieving the best average rank among 102 configurations in challenging metrics such as VUS-PR while exhibiting 40% lower training instability than its closest TS2I competitor.
- C2: We propose a TSAD-oriented preprocessing and evaluation workflow that: i) uses an ACF-driven procedure to select an appropriate window size, ii) employs autoencoder-based reconstruction with 3 vision backbones (CNN-AE, ResNet18-AE, PVT-v2-b1-AE) and 3 transfer learning strategies (frozen encoder, progressive unfreezing, and differential learning rates), and iii) supports point-wise anomaly prediction and scoring, improving robustness to windowing and stride choices.
- C3: We conduct extensive experiments (24,430 in total) on the TSB-AD benchmark (Liu et al., 2024), comparing SPIRAL against 9 state-of-the-art transformations, 3 vision backbones, and 32 time-domain baselines across 23 datasets and 9 metrics.
- C4: We perform rigorous statistical analysis using the Friedman test with post-hoc Nemenyi test, providing Critical Difference diagrams that confirm statistically significant performance differences across methods.
- C5: We analyze computational efficiency and training dynamics, demonstrating that SPIRAL achieves remarkably high throughput while providing superior convergence stability across diverse vision backbone configurations.

This paper is organized as follows: Sect. 2 presents related work, Sect. 3 introduces SPIRAL, Sect. 4 describes the experimental setup, Sect. 5 reports the results, and Sect. 6 concludes the paper.

2 Related Work

2.1 Time Series Anomaly Detection

Anomaly detection (AD) aims to identify data instances that exhibit deviations from expected or normal behavior (Pang et al., 2021). Such deviations often correspond to critical and actionable events embedded within large-scale datasets, making their detection of considerable importance. Consequently, AD has found widespread application across numerous domains, including finance (Mienye & Jere, 2024), cloud computing (Smendowski & Nawrocki, 2024), and IoT (Anand et al., 2026).

Time-series anomaly detection (TSAD), which extends AD to sequential data, addresses the challenge of discovering abnormal behavior by analyzing how observations evolve over time (Lu et al., 2023). TSAD must explicitly account for temporal structure, as abnormal events can manifest either as brief isolated irregularities (point-wise outliers) or as sustained deviations unfolding over consecutive time steps (pattern-wise outliers) (Zamanzadeh Darban et al., 2024). This challenge is further compounded by non-stationary environments and contextual anomalies characteristic of real-world settings (Faber et al., 2026). For this reason, most TSAD techniques transform the input time series into overlapping temporal segments using sliding-window techniques, allowing models to reason over localized temporal contexts (Faber et al., 2025). In this work, we pursue an alternative direction: representing time series as images and leveraging vision backbones for anomaly detection, systematically evaluating their performance against powerful time-domain baselines.

In TSAD, autoencoders (AEs) based on Long Short-Term Memory (LSTM) networks have become a reference due to their effectiveness in modeling long-term temporal dependencies and complex sequential patterns (Alfeo et al., 2020). Building upon standard LSTM, the extended LSTM (xLSTM) architecture enhances memory expressiveness and training stability (Faber et al., 2025). However, native time-domain approaches operate directly on one-dimensional temporal representations, potentially overlooking spatial patterns and structural relationships that become more apparent in alternative representation spaces. In this study, we focus on time series to image (TS2I) transformations, comparing nine state-of-the-art techniques and introducing SPIRAL, a novel transformation method, alongside a formalized workflow for vision-based TSAD. Following the autoencoder paradigm, we employ three vision backbones: CNN-AE, ResNet18-AE, and Pyramid Vision Transformer (PVT-v2-b1)-AE, combined with three transfer learning (TL) strategies to evaluate transferability from ImageNet-pretrained models, resulting in a multi-faceted evaluation suite covering 24,430 experiments.

TSAD operates in two settings: univariate, where a single temporal signal is monitored, and multivariate, where multiple interdependent variables are analyzed jointly (Lu et al., 2023). While multivariate TSAD captures cross-variable correlations relevant to complex systems such as industrial processes, univariate TSAD remains fundamentally important for scenarios involving single-sensor measurements or partially observed systems (Braei &

Wagner, 2020). Notably, the majority of state-of-the-art TS2I transformations were designed as general-purpose univariate methods. To ensure fair comparison and isolate the contribution of our transformation, we follow this convention and introduce SPIRAL, a novel TS2I transformation specialized for TSAD, featuring: i) parameter-free design requiring no hyperparameter tuning, ii) linear time complexity with respect to window length, iii) asymmetric spatial encoding that eliminates redundancy, and iv) anomaly-centric representation that amplifies deviations for vision-based detection.

TSAD evaluation has been limited by dataset quality issues, inconsistent anomaly labels, and lack of standardized protocols. To address these challenges, the TSB-AD benchmark Liu et al. (2024) was introduced as a consolidated evaluation suite unifying widely used datasets within a common evaluation protocol. In this study, we leverage TSB-AD-U, conducting experiments on 23 datasets from diverse domains, employing 9 evaluation metrics including VUS-PR and VUS-ROC (Boniol et al., 2025), and comparing vision-based TSAD against 32 time-domain baselines.

2.2 Time Series to Image Transformations

Transforming 1D signals into 2D images has emerged as an alternative paradigm for time-series analysis. The primary motivation for TS2I transformations is to encode temporal dynamics into spatial structures that can be effectively modeled by computer vision architectures such as convolutional neural networks (CNNs) (Ni et al., 2025). By leveraging spatial analysis and hierarchical feature extraction mechanisms inherent to CNNs, TS2I approaches aim to capture complex patterns that may be difficult to model directly in the time domain. Moreover, TS2I enables leveraging pre-trained backbones. While TS2I has shown promise in classification and forecasting, its application to TSAD has received less attention. Consequently, in this study, we propose SPIRAL, a novel TS2I transformation that prioritizes computational efficiency, asymmetric encoding, and explicit amplification of anomalous patterns.

Among the most straightforward TS2I transformations is the line plot (LP), which renders the time series as a two-dimensional visualization (Rodrigues et al., 2021; Yang et al., 2023). Despite its simplicity, LP can preserve salient temporal trends, abrupt changes, and structural irregularities in a form that convolutional architectures are well suited to exploit. Similarly simple is State Grid (SG), which performs a naive reshape of the 1D signal into a 2D image (Karami et al., 2025). However, both LP and SG often exhibit significant training instability when used with pre-trained vision backbones. In contrast, our proposed SPIRAL not only achieves high TSAD performance but also enables substantially improved convergence and training stability.

More structured TS2I transformations explicitly encode temporal dependencies through grounded mathematical formulations. Gramian Angular Summation Fields (GASF) and Gramian Angular Difference Fields (GADF) project time series into polar coordinates and compute cosine or sine-based angular interactions, embedding global temporal correlations into a dense image matrix (Wang & Oates, 2015). Markov Transition Fields (MTF) discretize values into quantile bins and represent first-order transition probabilities as a spatial field, emphasizing state dynamics (Wang et al., 2015). Recurrence Plots (RP) rely on phase-space reconstruction and pairwise distance thresholding to visualize recurring system states (Eckmann et al., 1995). However, these general-purpose transformations share com-

mon limitations: they often produce symmetric images with substantial redundancy, require hyperparameter tuning, and exhibit $O(N^2)$ computational complexity. SPIRAL addresses these shortcomings through asymmetric encoding, parameter-free design, and linear time complexity with respect to window length N .

A distinct subset of TS2I methods focuses on time-frequency representations (Percival & Walden, 2000). Ricker Wavelet Transform (RWT) and Morlet Wavelet Transform (MWT) map signals into joint time-frequency domains, capturing localized spectral characteristics through the Mexican hat wavelet (Lau & Weng, 1995) and continuous wavelet transformation with Morlet basis (Rhif et al., 2019). These transformations are particularly effective for detecting anomalies in signals with non-stationary frequency components. However, their emphasis on spectral decomposition may limit sensitivity to anomalies characterized by amplitude deviations. SPIRAL addresses this limitation by directly encoding signal amplitude through its spiral mapping, enabling effective detection of point-wise spikes. Spiral layouts have been used for data visualization to reveal periodic patterns (Weber et al., 2001). However, these approaches render data as graphical elements (lines, bars) along a spiral axis just for human inspection. In contrast, SPIRAL is the TS2I transformation that encodes time series as dense pixel-intensity images on a spiral geometry for machine learning (ML), with formal locality-preserving properties designed for TSAD.

Table 1 provides a structured overview of the TS2I transformations evaluated in this study. GASF, GADF, MTF, and RP rely on full $N \times N$ pairwise interaction matrices, yielding $O(N^2 + S^2)$ per-window cost and introducing sensitivity to method-specific hyperparameters such as normalization range, quantile bin count, or phase-space embedding configuration. LP, SG, RWT, and MWT avoid pairwise computation but were not designed with anomaly detection as their primary objective: LP and SG produce general-purpose visual representations, RWT and MWT prioritize spectral decomposition over amplitude deviations (Percival & Walden, 2000), and RN serves as a lower-bound control that generates

Table 1 Structured comparison of TS2I transformations evaluated in this study

Method	Category	Hyperparameters	Complexity	TSAD-specific
RN	Random noise	Parameter-free	$O(S^2)$	×
LP	Direct visualization	Parameter-free	$O(N + S^2)$	×
SG	Time series reshape	Parameter-free	$O(N + S^2)$	×
GASF	Angular field	Normalization range	$O(N^2 + S^2)$	×
GADF	Angular field	Normalization range	$O(N^2 + S^2)$	×
MTF	Transition field	Quantile bins, discretization	$O(N^2 + S^2)$	×
RP	Phase-space recurrence	Embedding dim., delay, threshold	$O(N^2 + S^2)$	×
RWT	Time-frequency	Scale range (frequency resolution)	$O(N \log N + S^2)^\dagger \times$	
MWT	Time-frequency	Center frequency, scale range (frequency resolution)	$O(N \log N + S^2)^\dagger \times$	

Complexity is reported per window of length N for a fixed output image size $S \times S$; the S^2 term is shared by all methods and accounts for image allocation and resizing. TSAD-specific indicates whether anomaly detection was the primary design objective

[†] Computed via FFT-based continuous wavelet transform; the $O(N \log N)$ factor applies per scale, with the number of scales determined by the configured scale range

images independently of the input signal. The TSAD-specific column confirms that no existing TS2I method was designed with anomaly detection as its primary objective, establishing the representational gap this work addresses.

All methods in Table 1 are designed for univariate input: each maps a single scalar-valued window to a 2D image and does not explicitly model dependencies across multiple variables. SG is the most directly extensible to multivariate settings, as its reshape operation is agnostic to the semantic meaning of the input axes; nevertheless, even this extension requires explicit decisions on channel composition that go beyond the core transformation design (Ni et al., 2025). We therefore follow the univariate convention of existing TS2I methods, which allows the contribution of the transformation itself to be evaluated independently of channel composition (channelization) strategy and cross-variable dependency modeling.

2.3 Summary

Recent advances in TSAD have been driven primarily by deep sequential models, with LSTM-based architectures and their extensions establishing strong baselines (Paparrizos et al., 2025). In parallel, TS2I transformations have demonstrated effectiveness in forecasting (Barra et al., 2020) and classification (Hatami et al., 2017), yet their application to TSAD remains less studied. Consequently, existing TS2I methods are general-purpose and designed to capture global temporal structure, not to amplify rare or subtle deviations characteristic of anomalies.

Current TS2I techniques exhibit four key limitations (Ni et al., 2025). First, they encode heterogeneous aspects of temporal structure making detection performance dependent on the specific anomaly type. Second, many transformations incur $O(N^2)$ computational complexity, produce symmetric images with redundancy, or require extensive hyperparameter tuning. Third, no existing transformation has been designed with TSAD as its primary use case. Fourth, several methods cause high training instability due to inconsistent representations, which is a limitation for continual and online learning.

This study addresses these gaps by proposing SPIRAL, a novel TS2I transformation specialized for TSAD. Following the univariate convention adopted by existing TS2I methods, we evaluate SPIRAL on the TSB-AD-U benchmark across 23 datasets and 9 metrics, comparing against 32 time-domain baselines. This rigorous evaluation isolates detection performance from cross-channel dependencies, establishing SPIRAL as a robust building block for vision-based TSAD.

3 SPIRAL

In this section, we introduce SPIRAL (Sequence Projection via Radial Arrangement for anomaly detection), a novel univariate TS2I transformation for TSAD.

3.1 Mathematical Formulation

Given a univariate time series window $\mathbf{w} = [w_1, w_2, \dots, w_W]$ of length W and image size $S \times S$, SPIRAL maps temporal indices to spatial coordinates via a two-arm Archimedean

spiral (Sasaki et al., 2020). Specifically, for each pixel at position (i, j) the radial distance from the image center can be defined as :

$$r_{i,j} = \sqrt{(i - S/2)^2 + (j - S/2)^2}. \quad (1)$$

Then, we compute the polar angle $\theta_{i,j}$ leveraging the *atan2* function (the 2-argument arctangent).

$$\theta_{i,j} = \text{atan2}(j - S/2, i - S/2). \quad (2)$$

Our rationale for choosing the *atan2* instead of *atan* is to give the correct angle for every pixel around the center, in all four quadrants. In contrast, *atan* yields values in the range $[-\pi/2, \pi/2]$, which does not allow to distinguish between points in opposite quadrants. However, since polar angles can have negative angles (*atan2* returns from values in range $[-\pi, \pi]$), we adjust it as:

$$\theta_{i,j} = \begin{cases} \theta_{i,j} + 2\pi & \text{if } \theta_{i,j} < 0 \\ \theta_{i,j} & \text{if } \theta_{i,j} \geq 0, \end{cases} \quad (3)$$

to ensure that spiral arms rotate in a consistent counter-clockwise direction from $[0, 2\pi]$, so that adjacent pixels correspond to the correct temporal order.

Subsequently, the two-arm spiral is unfolded by translating radial distance into angular progression, and creating the path along which the time series is painted into the 2D image:

$$\theta'_{i,j} = \theta_{i,j} + \frac{r_{i,j}}{S/2} \cdot 4\pi, \quad (4)$$

where 4π corresponds to the two-arm Archimedean spiral, and $r_{i,j}/(S/2)$ part scales the radial distance to $[0, 1]$, making the spiral wind gradually from center (0) to edge (1).

Finally, we map pixel (i, j) to a time series index. The transformation is applied within a circular region:

$$\text{id}x_{i,j} = \left\lfloor \frac{\theta'_{i,j}}{2\pi} \cdot \frac{W}{2} \right\rfloor \bmod W, \quad (5)$$

The transformation is then applied within a circular region:

$$I_{i,j} = \begin{cases} w_{\text{id}x_{i,j}} & \text{if } r_{i,j} \leq S/2 \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

This spiral mapping creates a spatially continuous representation where temporally adjacent points remain geometrically close. Unlike existing methods that prioritize preserving complete temporal information, SPIRAL focuses on encoding patterns that maximize anomaly discriminability.

We illustrate SPIRAL's encoding on a synthetic sinusoidal signal under 10 conditions: normal and 9 anomaly types. Figure 1 shows the raw series. In Fig. 2, where we illustrate

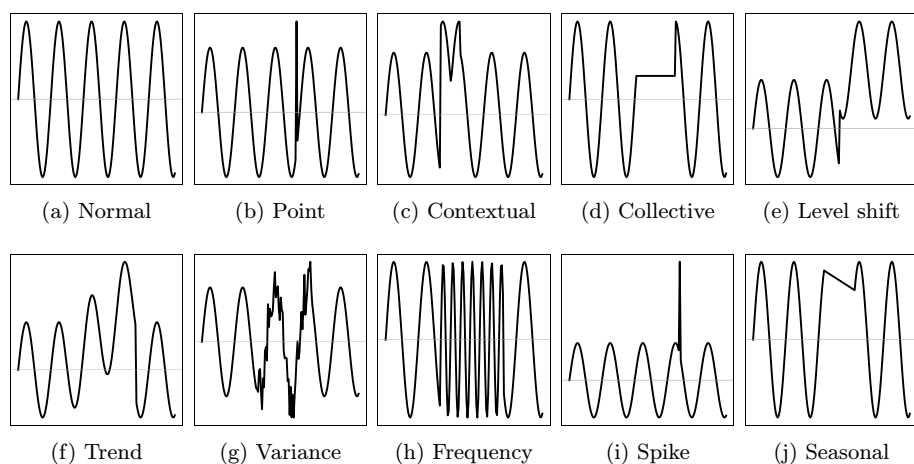


Fig. 1 Time series anomaly types in univariate signals

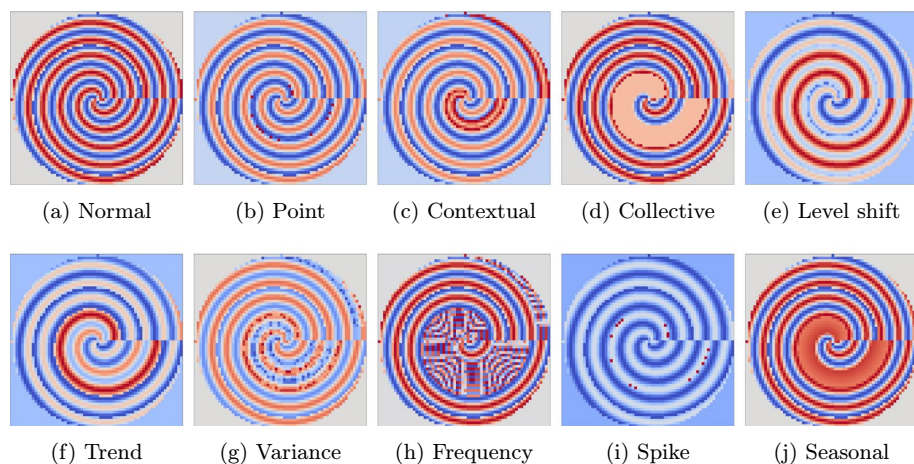


Fig. 2 SPIRAL transformation of time series anomaly types

SPIRAL images, temporal progression unfolds along spiral arms from the center outward, with intensity encoding amplitude. The normal signal shows smooth bands, while anomalies produce characteristic disruptions: spikes as isolated spots, level shifts as abrupt transitions, trends as gradual drifts, variance changes as contrast variation, frequency changes as band spacing shifts, and collective, contextual, or seasonal anomalies as extended or irregular patterns.

3.2 Core Principles

SPIRAL is guided by six core principles:

1. **Anomaly-centric encoding:** Instead of preserving complete signal fidelity, SPIRAL emphasizes encoding deviations and structural breaks that characterize anomalies, serving as an “alarm signal” for vision models.
2. **Parameter-free simplicity:** SPIRAL requires no hyperparameter tuning. In contrast, RP depends on the choice of embedding dimension, time delay, and distance threshold (Marwan et al., 2007); MTF requires specifying the number of quantile bins and discretization strategy (Wang et al., 2015); while GAF involves normalization range selection that affects the polar coordinate mapping. Prior work has shown that such parameter choices may overemphasize local patterns at the expense of global structure (Jiang et al., 2025), creating a substantial calibration difficulties.
3. **Linear complexity:** Unlike symmetric transforms that compute all pairwise relationships (e.g., GASF, GADF, RP) with $O(N^2 + S^2)$ complexity (the S^2 term corresponds to image resizing), SPIRAL requires $O(N + S^2)$ time. For fixed image size S , this results in linear scaling with respect to window length N .
4. **Asymmetric spatial encoding:** SPIRAL uses asymmetric mapping via spiral geometry, avoiding redundant information present in symmetric transforms where upper and lower triangular regions mirror each other.
5. **Model-agnostic:** As a pure data transformation, SPIRAL is independent of downstream models – neither learned nor fine-tuned during training, making it universally applicable across different architectures.
6. **Single-Channel Representation:** Existing TS2I methods produce single-channel images, typically replicated across RGB channels to maintain compatibility with pre-trained backbones. SPIRAL follows the same convention, deliberately adopting a single-channel design for two reasons. First, this ensures that performance improvements can be attributed to the transformation itself rather than increased channel capacity. Second, a single-channel representation remains universally applicable: it can be replicated for RGB backbones, used directly with grayscale-compatible architectures, or composed into multi-channel arrangements as needed. This design establishes SPIRAL as a model-agnostic building block for vision-based TSAD.

3.3 Rationale for Anomaly Detection

Traditional TS2I methods aim to preserve all temporal information for general-purpose classification or forecasting. In contrast, SPIRAL is designed to amplify anomaly-relevant patterns: the spiral mapping inherently emphasizes local changes (adjacent time points map to nearby pixels), while the radial dimension encodes temporal progression.

With SPIRAL, abrupt temporal deviations (spikes, shifts, local regime breaks) become localized geometric distortions in the image plane. This better matches the locality and hierarchical feature extraction bias of vision-based encoders, which are known to exploit edges, contours, and repeated motifs in 2D layouts (He et al., 2016; Wang et al., 2021). In other words, SPIRAL does not only convert a sequence into an image; it converts anomalies into forms that standard visual filters are structurally good at detecting.

This creates a representation where anomalies – characterized by abrupt changes or pattern breaks – manifest as visually distinctive structures that pretrained vision models can readily detect, even without task-specific fine-tuning (Fig. 2).

Several classical TS2I mappings (e.g., GASF, GADF, RP) encode broad temporal relationships, but often include substantial symmetric or near-symmetric structure. SPIRAL's asymmetric radial arrangement acts as a compact projection: each pixel is assigned a directional temporal role, reducing duplicated evidence while preserving anomaly-relevant discontinuities. A theoretical consequence is a higher anomaly-to-background contrast per unit area of the image, which can improve sample efficiency and optimization stability in downstream autoencoders.

By design, SPIRAL jointly encodes local neighborhoods and broader temporal progression in one coordinate system. This supports simultaneous sensitivity to micro-anomalies (isolated spikes) and macro anomalies (trend or variance shifts), without explicitly computing all pairwise dependencies. This is consistent with a multi-scale analysis perspective: useful anomaly evidence appears at different effective resolutions, and representations that expose these scales in a single visual field can be easier to model robustly.

3.4 Standardized TS2I Workflow for TSAD

We standardize the TS2I anomaly-detection pipeline for univariate time series by aligning preprocessing, training, and evaluation with point-wise labels while keeping windowing data-driven. Let $\mathbf{x} = [x_1, \dots, x_T]$ denote a univariate series with point-wise labels $\mathbf{y} = [y_1, \dots, y_T]$, $y_t \in \{0, 1\}$, and let $\mathbf{x}^{\text{tr}}$ be the training split. We compute the sample autocorrelation function (ACF) on $\mathbf{x}^{\text{tr}}$:

$$\rho(k) = \frac{\sum_{t=k+1}^{T_{\text{tr}}} (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^{T_{\text{tr}}} (x_t - \bar{x})^2}, \quad k = 1, \dots, K, \quad (7)$$

and we select the window length W as the first lag at which the sample ACF falls below the 95% confidence bound under the white-noise null hypothesis:

$$W = \min\{k \geq 1 : |\rho(k)| < \tau\}, \quad \tau = \frac{1.96}{\sqrt{T_{\text{tr}}}}, \quad (8)$$

where τ is the 95% confidence bound arising from Bartlett's theorem (1946): under $H_0 : \rho(k) = 0$, the sample autocorrelation $\rho(k)$ is asymptotically normal with variance $1/T_{\text{tr}}$, and 1.96 is the 97.5th percentile of the standard normal distribution, giving a two-sided 95% critical value (Hwang & Vogelsang, 2026). The rule is applied to the training split only, ensuring that test-set information does not influence windowing. The ACF is computed up to $K_{\text{max}} = \min(100, \lfloor T_{\text{tr}}/4 \rfloor)$ lags; if no lag in $[1, K_{\text{max}}]$ satisfies Eq. (8), W is set to $\lfloor K_{\text{max}}/2 \rfloor$ as a fallback. This produces windows that reflect the dominant temporal structure in each series.

Given W , we segment train and test splits separately into contiguous, non-overlapping windows with stride W :

$$\mathbf{w}^{(m)} = [x_{(m-1)W+1}, \dots, x_{mW}], \quad m = 1, \dots, M = \lfloor T/W \rfloor. \quad (9)$$

When T is not divisible by W , the final incomplete stride is discarded. This choice is preferred over zero-padding, which would introduce an artificial constant region into the final TS2I image, potentially creating a systematic reconstruction bias for that window. Discarding the final partial segment avoids this artifact entirely; the at most $W - 1$ tail points inherit the reconstruction score of the last complete window, which constitutes a negligible boundary effect for the series lengths in TSB-AD-U.

The corresponding window label is defined by OR-pooling of point labels,

$$\tilde{y}^{(m)} = \max_{t \in \mathcal{J}_m} y_t, \quad \mathcal{J}_m = \{(m-1)W + 1, \dots, mW\}, \quad (10)$$

so a window is anomalous if it contains at least one anomalous point.

Each window is then mapped to a single-channel image using a TS2I transform $\mathcal{T}$ (e.g., SPIRAL),

$$I^{(m)} = \mathcal{T}(\mathbf{w}^{(m)}) \in \mathbb{R}^{S \times S}, \quad (11)$$

and replicated to three channels for compatibility with pretrained vision backbones. An autoencoder f_θ is trained to reconstruct $I^{(m)}$, producing $\hat{I}^{(m)}$, and the window anomaly score is the reconstruction error:

$$e^{(m)} = \frac{1}{S^2} \left\| I^{(m)} - \hat{I}^{(m)} \right\|_2^2. \quad (12)$$

To align with point-wise evaluation in TSB-AD, we propagate the window score to all points inside that window:

$$s_t = e^{(m)} \quad \forall t \in \mathcal{J}_m. \quad (13)$$

This yields a point-wise score vector $\mathbf{s}$ that can be directly compared to the benchmark labels $\mathbf{y}$ without altering their resolution. In the case of a mismatch between the length of propagated scores and the ground truth labels, the remaining tail points inherit the reconstruction score of the last complete window to achieve the same length.

4 Experimental Setup

In our experiments, performed on NVIDIA GH200 GPUs, we leverage the TSB-AD benchmark and follow its standardized evaluation protocol (Liu et al., 2024). For reproducibility, we share our code publicly at <https://github.com/Smendowski/SPIRAL>. The experimental suite comprises 23 datasets, 10 TS2I transformation methods, and 7 autoencoder models, yielding a total of 24,430 experiments. We compare our TS2I-based approaches against 32 established time-domain baselines across 9 evaluation metrics.

4.1 Datasets

The TSB-AD benchmark (Liu et al., 2024) includes a curated collection of 23 univariate and 20 multivariate publicly available time-series datasets. These datasets span diverse real-world domains: medical monitoring (LTDB, MITDB, SED, SVDB), aerospace sensors (MSL, SMAP), server and web service metrics (IOPS, NAB, NEK, WSD, YAHOO), industrial facilities (Exathlon, Power, SMD, SWaT), human activity recognition (Daphnet, OPPORTUNITY), network traffic (CATSv2), environmental monitoring (TAO), financial markets (Stock), and synthetic anomalies (MGAB, TODS, UCR). This diversity ensures robust evaluation across varied anomaly characteristics, temporal patterns, and application contexts. Detailed descriptions of individual datasets are available in the original TSB-AD benchmark paper (Liu et al., 2024), and all datasets are publicly accessible at <https://github.com/TheDatumOrg/TSB-AD>.

4.2 Metrics

We evaluate TSAD performance using nine metrics spanning three categories. First, AUC-ROC and AUC-PR measure threshold-independent classification performance, with AUC-PR being particularly informative for imbalanced anomaly detection tasks. Second, VUS-ROC and VUS-PR (Paparrizos et al., 2022) extend traditional AUC metrics to account for temporal segments, rewarding early detection within anomaly ranges. Third, Standard-F1, PA-F1 (Tatbul et al., 2018), Event-F1 (Hwang et al., 2019), R-F1 (Tatbul et al., 2018), and Affiliation-F1 (Huet et al., 2022) employ different strategies for matching predicted and ground truth anomaly events, ranging from point-wise evaluation to segment-overlap requirements. For statistical comparison, we use the Friedman-Nemenyi test ($\alpha = 0.05$) to rank methods across datasets, with Critical Difference indicating statistically significant performance differences (Demšar, 2006).

4.3 Baseline TS2I Transformations

We compare SPIRAL against nine established TS2I transformations:

- **Random Noise (RN):** Gaussian noise serving as a lower-bound control, testing whether models learn from signal structure or spurious patterns.
- **State Grid (SG):** Naive 1D-to-2D resizing of a time series window into a square image. This encodes temporal patterns only through positional mapping, without computing pairwise interactions or other transformations.
- **Line Plot (LP):** Direct pixel-wise rendering of the time series as a line graph, preserving raw temporal structure without transformation.
- **Gramian Angular Summation Field (GASF)** (Wang & Oates, 2015): Polar coordinate transformation encoding temporal correlations as cosine summation of angular differences.
- **Gramian Angular Difference Field (GADF)** (Wang & Oates, 2015): Variant of GASF using sine differences to capture complementary temporal relationships.
- **Markov Transition Field (MTF)** (Wang et al., 2015): Discretizes time series into quantile bins and encodes first-order temporal transitions as a probabilistic field.

- **Recurrence Plot (RP)** (Eckmann et al., 1995): Phase-space reconstruction visualizing recurrent states through pairwise distance thresholding in embedded dimensions.
- **Ricker Wavelet Transform (RWT)**: Time-frequency decomposition using the Mexican hat wavelet, capturing localized frequency characteristics.
- **Morlet Wavelet Transform (MWT)**: Continuous wavelet transform with Morlet basis, providing time-frequency localization with Gaussian windowing.

These methods represent complementary approaches: direct visualization (LP, SG), Gramian angular fields (GASF, GADF), phase-space dynamics (RP), transition probabilities (MTF), and time-frequency decomposition (RWT, MWT). For fairness, all transformation approaches follow the same window-size optimization workflow based on ACF described in Sect. 3.4. Additional visualizations showing how each baseline TS2I transformation encodes different anomaly types from Fig. 1 are provided in online Appendix A. Note that images are displayed using colormaps for enhanced interpretability; however, all transformations produce single-channel grayscale representations that are replicated across RGB channels for compatibility with pretrained vision models. For all experiments in this study, the image size S is set to 64, producing images with a resolution of 64×64 .

4.4 Baseline Time-Domain Models

We compare our TS2I-based approaches against 32 established time-domain anomaly detection methods from the TSB-AD benchmark (Liu et al., 2024). These baselines span diverse methodological paradigms including statistical methods (Sub-PCA, Sub-LOF, Sub-MCD), distance-based approaches (KShapeAD, MatrixProfile, Sub-KNN), deep learning models (LSTM-AD, AutoEncoder, TranAD, OmniAnomaly, AnomalyTransformer), foundation models (MOMENT, TimesFM, Lag-Llama, Chronos, OFA), and ensemble methods (IForest, Sub-IForest). All baseline models use identical hyperparameters and configurations as specified in the TSB-AD benchmark to ensure fair comparison. Detailed model descriptions and parameter settings are provided in the TSB-AD paper (Liu et al., 2024).

4.5 Vision-Domain Models

In this study, we employ autoencoder-based models for TSAD on univariate TS2I representations. Autoencoders are widely used for anomaly detection due to their ability to learn compressed representations of normal patterns and detect deviations via reconstruction error (Pinaya et al., 2020). We compare three encoder architectures:

- **CNN-AE**: A lightweight convolutional autoencoder trained from scratch.
- **ResNet18-AE**: A convolutional autoencoder based on the ResNet18 backbone (He et al., 2016), pretrained on ImageNet.
- **PVT-v2-b1-AE**: An autoencoder using Pyramid Vision Transformer (PVT-v2-b1 variant) (Wang et al., 2021), pretrained on ImageNet, employing hierarchical self-attention.

Both ResNet18-AE and PVT-v2-b1-AE share an identical convolutional decoder (1.9M trainable parameters), enabling fair comparison by isolating the impact of encoder design. For pretrained models, we evaluate three TL strategies: Frozen Encoder (FE), Progressive

Unfreezing (PU), and Differential Learning Rates (DLR). Detailed architectural specifications and hyperparameters are provided in online Appendix B.

5 Results Discussion

This research addresses the following research questions:

- **RQ1:** Can TS2I-based vision models with SPIRAL compete with state-of-the-art time-domain anomaly detection methods on univariate time series?
- **RQ2:** How do TS2I transformations compare in computational efficiency, and does SPIRAL's expressive representation come at a higher computational cost?
- **RQ3:** Which TS2I transformation yields the best anomaly detection performance, and does SPIRAL outperform existing alternatives?
- **RQ4:** How do training dynamics (convergence speed, stability, and cost) vary across TS2I transformations, and does SPIRAL's performance come at the cost of slower or less stable training?
- **RQ5:** Which vision backbone architecture and transfer learning strategy maximizes anomaly detection performance in the TS2I paradigm?

5.1 TS2I-Based Vision Backbones vs. Time-Domain Baselines

Table 2 reports mean TSAD across all nine evaluation metrics for all 102 methods. Fine-grained per-metric tables are provided in online Appendix E. To streamline the discussion, we focus on VUS-PR as the primary metric. We choose VUS-PR for two reasons: first, it extends precision-recall evaluation – which is more informative than ROC for imbalanced TSAD (Paparrizos et al., 2022); second, our point-wise scoring strategy propagates window-level scores to individual time points, making VUS-PR a rigorous measure that penalizes false alarms and detection delays at fine temporal granularity.

Overall ranking. We first applied the Friedman test to verify whether statistically significant differences exist among the 102 methods across all 23 datasets. The null hypothesis was rejected ($p < 0.05$), confirming that meaningful performance differences exist. We then conducted a post-hoc Nemenyi test ($\alpha = 0.05$). PVT-v2-b1-AE-PU (SPIRAL) achieves the lowest average rank of 23.87 in VUS-PR, placing it first among all 102 evaluated methods. Notably, all 15 top-ranked methods are TS2I-based configurations, demonstrating that the vision-based paradigm as a whole is highly competitive. The best-performing time-domain baseline, KShapeAD, has an average rank of 34.26 – more than 10 positions behind SPIRAL. A graphical critical difference plot for VUS-PR showcasing the top 20 methods is shown in Fig. 3. Remaining per-metric diagrams are provided in online Appendix E. While this gap falls within the critical difference threshold ($CD = 24.87$), the dominance of TS2I-based methods across all top 20 positions suggests a systematic advantage of the proposed SPIRAL method over time-domain approaches. This difference becomes statistically significant for majority of comparisons in the full analysis involving 102 methods. Due to space limitations, we direct the interested reader to the full comparison (102 methods) in our online Appendix (see Fig. 3).

Table 2 Mean performance across 9 metrics for TS2I-based vision backbones and 32 time-domain baselines on 23 datasets of TSB-AD-U

Method	AUC-PR	AUC-ROC	VUS-PR	VUS-ROC	Stan- dard- F1	PA- F1	Event- based- F1	R- based- F1	Af- filia- tion- F1
CNN-AE (RN)	0.07	0.54	0.13	0.59	0.14	0.26	0.15	0.14	0.74
CNN-AE (LP)	0.27	0.72	0.36	0.76	0.34	0.56	0.44	0.31	0.83
CNN-AE (SG)	0.33	0.76	0.41	0.80	0.39	0.60	0.51	0.36	0.85
CNN-AE (GASF)	0.17	0.67	0.28	0.72	0.25	0.52	0.35	0.25	0.81
CNN-AE (GADF)	0.13	0.55	0.20	0.63	0.20	0.43	0.26	0.20	0.76
CNN-AE (MTF)	0.15	0.59	0.22	0.66	0.21	0.46	0.29	0.21	0.77
CNN-AE (RP)	0.15	0.60	0.20	0.65	0.21	0.38	0.25	0.20	0.70
CNN-AE (RWT)	0.24	0.72	0.32	0.77	0.32	0.57	0.44	0.28	0.83
CNN-AE (MWT)	0.23	0.71	0.32	0.76	0.31	0.57	0.44	0.29	0.84
CNN-AE (SPIRAL)	0.32	0.75	0.41	0.79	0.39	0.60	0.52	0.37	0.85
ResNet18-AE (RN)	0.08	0.54	0.14	0.59	0.14	0.27	0.16	0.16	0.75
ResNet18-AE (LP)	0.25	0.69	0.35	0.74	0.32	0.55	0.42	0.30	0.84
ResNet18-AE (SG)	0.34	0.73	0.40	0.79	0.40	0.60	0.52	0.37	0.85
ResNet18-AE (GASF)	0.17	0.67	0.28	0.73	0.26	0.54	0.35	0.26	0.81
ResNet18-AE (GADF)	0.14	0.55	0.20	0.63	0.21	0.46	0.29	0.22	0.77
ResNet18-AE (MTF)	0.15	0.59	0.21	0.66	0.21	0.46	0.28	0.22	0.77
ResNet18-AE (RP)	0.15	0.58	0.21	0.64	0.22	0.40	0.28	0.22	0.74
ResNet18-AE (RWT)	0.24	0.72	0.31	0.77	0.31	0.58	0.44	0.29	0.83
ResNet18-AE (MWT)	0.21	0.70	0.30	0.75	0.29	0.56	0.43	0.28	0.83
ResNet18-AE (SPIRAL)	<u>0.36</u>	0.76	<u>0.43</u>	0.81	0.42	0.63	0.55	0.39	0.86
PVT-v2-b1-AE (RN)	0.08	0.53	0.13	0.58	0.14	0.27	0.15	0.16	0.75
PVT-v2-b1-AE (LP)	0.26	0.70	0.37	0.74	0.34	0.57	0.45	0.32	0.85
PVT-v2-b1-AE (SG)	0.35	0.75	0.41	0.80	0.41	0.61	0.53	0.38	0.85

Table 2 (continued)

Method	AUC-PR	AUC-ROC	VUS-PR	VUS-ROC	Stan- dard- F1	PA- F1	Event- based- F1	R- based- F1	Af- filia- tion- F1
PVT-v2-b1-AE (GASF)	0.18	0.68	0.30	0.74	0.26	0.55	0.37	0.26	0.82
PVT-v2-b1-AE (GADF)	0.14	0.55	0.20	0.63	0.21	0.48	0.30	0.22	0.77
PVT-v2-b1-AE (MTF)	0.15	0.59	0.22	0.66	0.22	0.47	0.29	0.22	0.78
PVT-v2-b1-AE (RP)	0.17	0.58	0.22	0.65	0.23	0.41	0.28	0.23	0.74
PVT-v2-b1-AE (RWT)	0.24	0.73	0.31	0.78	0.32	0.57	0.44	0.28	0.83
PVT-v2-b1-AE (MWT)	0.22	0.71	0.31	0.76	0.29	0.56	0.43	0.28	0.83
PVT-v2-b1-AE (SPIRAL)	0.35	<u>0.76</u>	0.43	<u>0.81</u>	<u>0.42</u>	0.63	0.55	0.39	0.86
Sub-PCA	0.37	0.71	0.42	0.76	0.42	0.56	0.49	0.41	0.85
KShapeAD	0.35	0.74	0.40	0.76	0.39	0.58	0.46	<u>0.40</u>	0.83
POLY	0.31	0.73	0.39	0.76	0.37	0.53	0.45	0.35	0.85
Series-2Graph	0.33	0.76	0.39	0.80	0.38	0.65	0.50	0.35	0.85
MOMENT (FT)	0.30	0.69	0.39	0.76	0.35	0.65	0.49	0.35	0.86
MOMENT (ZS)	0.30	0.68	0.38	0.75	0.35	0.61	0.49	0.36	0.86
KMeansAD	0.32	0.74	0.37	0.76	0.37	0.56	0.44	0.38	0.82
USAD	0.32	0.66	0.36	0.71	0.37	0.50	0.43	0.40	0.84
Sub-KNN	0.27	0.76	0.35	0.79	0.34	0.61	0.43	0.32	0.84
MatrixProfile	0.26	0.73	0.35	0.76	0.33	0.63	0.44	0.32	0.84
SAND	0.29	0.73	0.34	0.76	0.35	0.56	0.42	0.36	0.81
CNN	0.33	0.71	0.34	0.79	0.38	0.78	<u>0.66</u>	0.35	0.88
LSTMAD	0.31	0.68	0.33	0.76	0.37	0.71	0.59	0.34	0.86
SR	0.32	0.74	0.32	0.81	0.38	0.87	0.67	0.35	0.89
TimesFM	0.28	0.67	0.30	0.74	0.34	<u>0.84</u>	0.63	0.34	<u>0.89</u>
IForest	0.29	0.71	0.30	0.78	0.35	0.73	0.56	0.30	0.84
OmniAnomaly	0.27	0.65	0.29	0.72	0.31	0.59	0.46	0.29	0.83
Lag-Llama	0.25	0.65	0.27	0.72	0.30	0.77	0.59	0.31	0.88
Chronos	0.26	0.66	0.27	0.73	0.32	0.83	0.61	0.33	0.88
TimesNet	0.18	0.61	0.26	0.72	0.24	0.67	0.47	0.21	0.86
AutoEncoder	0.19	0.63	0.26	0.69	0.25	0.54	0.36	0.28	0.82
TranAD	0.20	0.57	0.26	0.68	0.25	0.58	0.43	0.25	0.83
FITS	0.17	0.61	0.26	0.73	0.23	0.65	0.42	0.20	0.86
Sub-LOF	0.16	0.68	0.25	0.73	0.24	0.57	0.35	0.25	0.82
OFA	0.16	0.59	0.24	0.71	0.22	0.67	0.45	0.20	0.86
Sub-MCD	0.15	0.67	0.24	0.72	0.23	0.54	0.32	0.24	0.81

Table 2 (continued)

Method	AUC-PR	AUC-ROC	VUS-PR	VUS-ROC	Standard-F1	PA-F1	Event-based-F1	R-based-F1	Affiliation-F1
Sub-HBOS	0.18	0.61	0.23	0.67	0.23	0.60	0.35	0.27	0.79
Sub-OCS-VM	0.16	0.65	0.23	0.73	0.22	0.55	0.32	0.23	0.79
Sub-IForest	0.16	0.63	0.22	0.72	0.22	0.63	0.34	0.23	0.80
Donut	0.14	0.56	0.20	0.68	0.20	0.57	0.38	0.20	0.82
LOF	0.14	0.58	0.17	0.68	0.21	0.62	0.41	0.22	0.79
AnomalyT-transformer	0.08	0.50	0.12	0.56	0.12	0.53	0.34	0.14	0.77

For ResNet18-AE and PVT-v2-b1-AE, results report the best-performing transfer learning strategy (FE, PU, or DLR) per backbone-TS2I combination. Best results in bold, second-best underlined. Full results for each transfer learning strategy are provided in online Appendix D

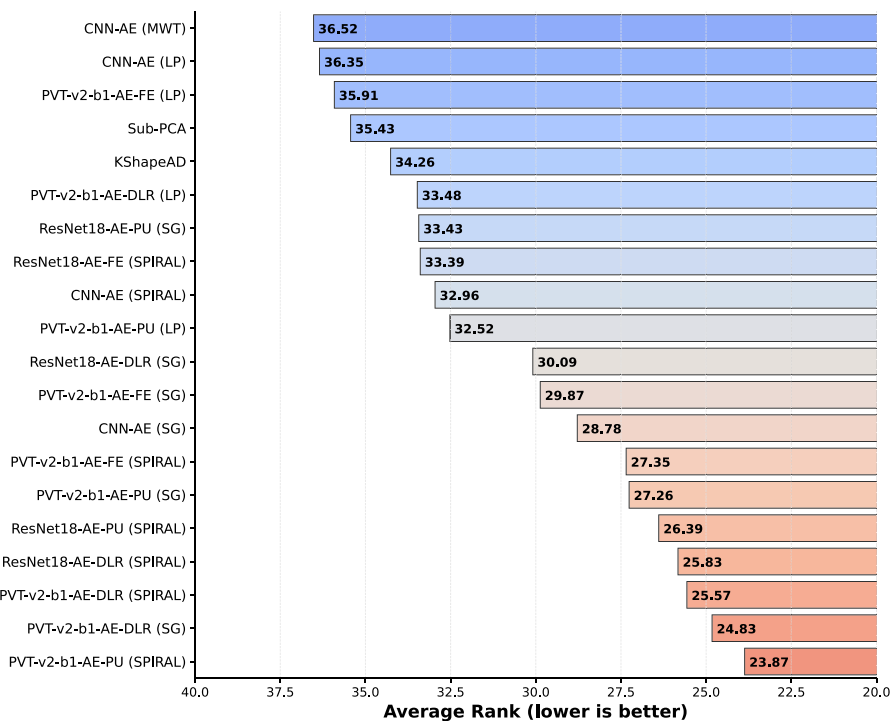


Fig. 3 Critical Difference diagram for VUS-PR limited to 20 highest ranked methods. Rankings determined via Friedman-Nemenyi test ($\alpha = 0.05$, $CD=24.87$, $k = 102$ methods, $N = 23$ datasets). For the full comparison, see online Appendix E

Stability vs. peak performance. A closer inspection of Table 10 reveals a pattern that explains this ranking. Several time-domain baselines achieve top performance on individual datasets but exhibit large swings across the benchmark. KShapeAD achieves the best VUS-PR on MITDB (0.69), SVDB (0.82), and LTDB (0.83), yet collapses to near-zero on MGAB

(0.02). Sub-PCA leads on OPP (0.91) but scores only 0.03 on SED. This reflects a common drawback: strong inductive biases that work well for specific signal characteristics but generalize poorly across diverse domains. In contrast, TS2I-based TSAD (especially SPIRAL-based configurations) maintain consistently competitive performance across all 23 datasets, never dropping below 0.06 on any single dataset, while achieving strong results where it excels – 0.95 on Exathlon, 0.88 on NEK, and 0.62 on SMD. This cross-domain consistency is precisely what drives the lowest average rank in the critical difference diagram.

Answering our **RQ1**, TS2I-based vision models not only compete with state-of-the-art time-domain methods but surpass them in overall ranking on the TSB-AD-U benchmark. SPIRAL achieves the best average rank across all 102 evaluated methods in VUS-PR. Crucially, this superiority stems not from dominating individual datasets but from a fundamentally different performance profile: where time-domain methods trade cross-domain consistency for specialised peak performance, SPIRAL delivers reliable anomaly detection across diverse time series characteristics without any domain-specific tuning. This is important for domains such as manufacturing, healthcare and cloud computing where stability of performance is critical (Smendowski & Nawrocki, 2024; Nawrocki & Smendowski, 2024). SPIRAL outperforms SG, the closest competitor, across all nine evaluation metrics by +1.2 to +4.9 percentage points. In addition, these gains come with 40% higher training stability (Sect. 5.3), making SPIRAL substantially more reliable for real-world deployment. Therefore, SPIRAL provides the greatest versatility and robustness across different datasets, overcoming the limitation of other transformations and time-domain baselines that perform reliably only on specific datasets.

Metrics where SPIRAL does not lead. SPIRAL does not rank first on all metrics. On PA-F1 and Event-based-F1, SR achieves 0.87 and 0.67 respectively, with CNN (0.78, 0.66) and LSTMAD (0.71, 0.59) also ranking substantially higher than the best SPIRAL configuration (0.63, 0.55). This gap reflects a structural consequence of the windowed scoring design. The TS2I workflow assigns a single reconstruction error to each non-overlapping window of length W and propagates it uniformly to all constituent time points; detection boundaries are therefore aligned with window edges rather than true anomaly onsets or off-sets. PA-F1 and Event-based-F1 reward methods that localize segment boundaries precisely, which naturally favors methods operating at point-level granularity such as SR and CNN. Affiliation-F1, which uses a distance-based matching criterion more tolerant of temporal boundary shifts, shows a correspondingly smaller gap: 0.86 for SPIRAL versus 0.89 for SR. However, we argue that VUS-PR, being threshold-independent and range-aware, is well-suited to the uniform windowed scoring strategy and is therefore the appropriate primary metric for this evaluation.

5.2 TS2I Transformation Efficiency

This section evaluates the computational efficiency of the TS2I transformation stage. This cost is incurred once per dataset during data preparation and is independent of the downstream model architecture or training procedure. Each time series is segmented into fixed-length, non-overlapping windows of length W determined adaptively via the Auto-Correlation Function (ACF). As shown in online Appendix C, window lengths vary substantially across domains – from $W = 10$ for aperiodic financial series (Stock, TAO) to $W = 58$ for complex oscillatory signals (CATSv2) – confirming that window length is inherently

data-driven. Each window is then mapped to a $3 \times S \times S$ image via one of ten TS2I transformations, with output replicated across channels for RGB backbone compatibility.

Table 3 and Fig. 4 reveal a clear efficiency hierarchy. Gramian-based methods (GASF, GADF, MTF) are the slowest, with throughput collapsing to 6–14 img/s on large-window datasets (Daphnet, Power) due to $\mathcal{O}(N^2 + S^2)$ pairwise computation – up to 55× slower than SG on the same datasets. LP, despite its conceptual simplicity, also ranks among the slowest (139–208 img/s) as pixel-level line rendering incurs significant per-sample overhead. Wavelets (RWT, MWT) offer stable intermediate throughput of 180–315 img/s across all datasets.

SG achieves the highest raw speed via trivial array reshaping (298–560 img/s), yet SPIRAL ranks second overall (average rank 2.04 vs. 1.09 for SG), remaining within 5–8% of SG across all 23 datasets (e.g., 521 vs. 560 img/s on NEK, 296 vs. 326 img/s on Daphnet) thanks to its fully vectorised $\mathcal{O}(N + S^2)$ radial index mapping. The CD diagram confirms SG and SPIRAL are statistically indistinguishable in throughput, while both significantly outperform all remaining methods ($\alpha = 0.05$, $CD = 2.54$). This answers our **RQ2**: a more expressive transformation does not imply higher cost – SPIRAL encodes geometrically rich spiral structure at near-identical speed to naive resizing. Moreover, while SG matches SPIRAL in transformation throughput, it exhibits 40% higher training instability (see Sect. 5.3), which can hinder deployment in continual or online learning scenarios where stable updates are essential. Thus, SPIRAL offers more favorable overall trade-off between efficiency, accuracy, and training stability.

5.3 Training Dynamics of TS2I-Based Vision Backbones

To streamline the discussion, in this section we also focus on VUS-PR as the primary evaluation metric. Consequently, Fig. 5 summarizes training dynamics versus TSAD performance across considered TS2I transformations, aggregated over all backbone-dataset combinations. Each point represents the mean VUS-PR and mean convergence speed – measured as the number of epochs required to reduce the training loss by 90% of its total observed decrease – averaged across all 23 datasets; error bars denote ± 1 standard error of mean (SEM). Bubble size is proportional to training instability, quantified as the coefficient of variation (CoV) of training loss over the final 20% of training epochs, averaged across all backbone-dataset combinations – smaller bubbles indicate more stable convergence.

The CoV is a standard measure of relative dispersion; its application as a late-stage training stability diagnostic for TS2I representations is introduced in this work. This metric is particularly relevant in continual and online learning settings, where stable convergence is critical: high loss variance in the final training stage may indicate susceptibility to catastrophic forgetting, while slow convergence may prevent the model from adapting before the next data batch arrives (Faber et al., 2026).

SPIRAL achieves the highest mean VUS-PR (0.421, SEM 0.058) while converging in 30.5 epochs (SEM 3.2) with moderate stability (CoV 0.123). The closest competitor, SG (VUS-PR 0.409), converges 19% faster (24.6 epochs) but exhibits 40% higher instability (CoV 0.172), consistent with its representational simplicity. A Friedman test across all ten TS2I transformations confirms that CoV values differ significantly across methods ($\chi^2 = 86.72$, $p < 0.001$, $n = 23$ datasets); a paired Wilcoxon signed-rank test further confirms that SPIRAL's CoV is significantly lower than SG's ($W = 14$, $p < 0.001$, one-sided). Wavelet-based transformations (RWT, MWT) offer competitive convergence (24 and 30

Table 3 Mean TS2I transformation throughput (images/second)

TS2I	CATSv2	Daphnet	Exathlon	IOPS	LTDB	MGAB	MITDB	MSL	NAB	NEK	OPP	Power
RN	338.81	282.69	294.57	316.41	318.91	267.02	300.20	408.00	330.31	416.76	276.14	303.17
LP	174.59	158.56	159.33	148.61	163.78	139.54	148.83	193.38	168.44	208.12	155.99	146.99
SG	422.82	325.66	354.60	367.10	395.45	297.47	375.51	526.66	374.36	560.03	327.90	365.69
GASF	45.23	6.13	19.95	256.28	216.41	14.63	98.68	300.48	256.38	313.34	219.39	10.00
GADF	45.31	6.37	19.95	255.88	217.59	14.63	100.46	350.75	260.25	345.44	227.77	10.01
MTF	70.71	7.99	26.51	260.17	246.47	23.40	121.92	387.23	271.98	413.34	227.04	14.09
RP	352.06	297.36	327.79	306.72	351.42	273.78	317.24	443.56	341.75	429.13	279.17	307.28
RWT	250.14	231.26	242.69	229.63	252.69	197.20	234.09	241.68	251.37	305.80	210.50	237.22
MWT	249.78	227.71	233.87	228.85	247.22	195.67	233.77	313.28	258.19	337.87	204.37	233.89
SPIRAL	350.01	296.45	344.34	329.63	363.89	288.35	324.69	492.10	365.57	521.83	306.22	305.92
TS2I	SED	SMAP	SMD	SVDB	SWaT	Stock	TAO	TODS	UCR	WSD	YAHOO	
RN	315.26	267.50	318.35	308.75	334.33	311.81	295.70	245.43	298.23	245.57	323.27	
LP	165.71	153.55	183.87	161.21	172.87	164.36	167.65	138.09	164.14	140.33	167.68	
SG	382.65	309.62	376.40	393.28	427.65	387.27	377.71	276.72	348.35	290.62	411.93	
GASF	9.63	240.70	181.71	61.60	65.00	273.81	275.59	197.05	96.04	201.17	256.39	
GADF	9.62	247.64	182.46	61.71	65.11	271.63	254.74	190.16	94.13	200.34	255.56	
MTF	14.96	240.28	173.44	74.50	77.34	294.64	282.69	239.55	78.69	213.72	292.39	
RP	320.54	285.51	340.89	345.75	381.76	350.87	339.74	242.24	308.16	247.89	335.24	
RWT	251.16	220.94	261.87	253.54	242.30	253.63	254.14	196.06	241.42	178.17	247.55	
MWT	211.79	215.04	255.34	242.79	249.06	249.53	254.17	191.81	234.64	185.78	258.16	
SPIRAL	372.70	310.22	341.02	348.77	411.48	364.36	384.05	254.94	330.03	258.18	392.54	

Higher values indicate faster transformation speed. Best results in bold, second-best underlined

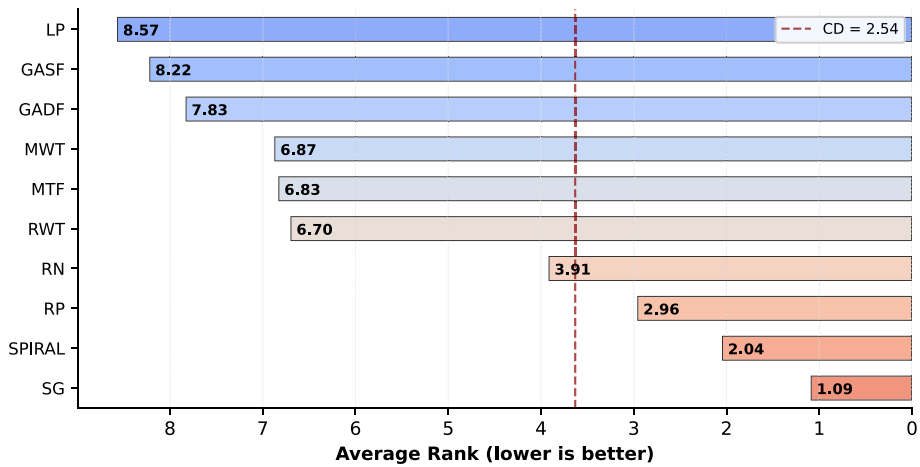


Fig. 4 Critical Difference diagram for TS2I transformation throughput (images/second) averaged across all datasets. Lower average rank indicates higher throughput (faster data processing). The SPIRAL, highlighted among top-ranked methods, consistently achieves high throughput while providing more sophisticated 2D representations compared to the naive SG (1D-to-2D resizing) approach

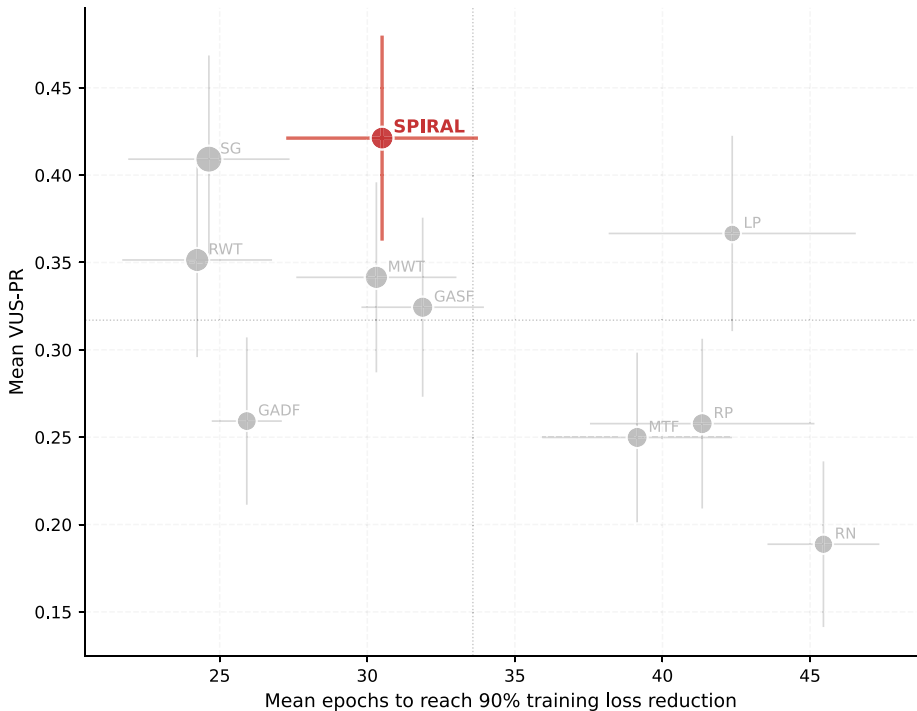
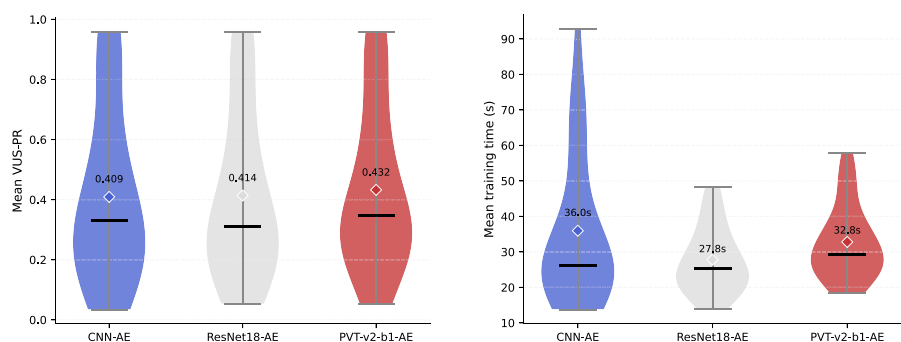


Fig. 5 Training dynamics vs. TSAD performance for all ten TS2I transformations. The x-axis reports mean epochs to 90% training loss reduction (lower is faster); the y-axis reports mean VUS-PR (higher is better). Bubble size is proportional to training instability (CoV of the loss in the final training stage) – smaller bubbles indicate more stable convergence. Dashed lines mark global means across all TS2I transformations

epochs) but fall 7–8 percentage points short of SPIRAL in VUS-PR. GASF, GADF, MTF, and RP combine slow convergence (32–45 epochs) with low VUS-PR (0.250–0.324). The naive baseline (RN) achieves a mean VUS-PR of only 0.189 at the slowest convergence (45.5 epochs), confirming that structured spatial content in the input image is a prerequisite for effective vision-based TSAD. This provides a clear answer to ours **RQ3**: SPIRAL yields the best TSAD performance among all evaluated TS2I transformations and enables stable training of vision backbones. This result is consistent across all eight remaining evaluation metrics, as confirmed by the analogous bubble charts provided in online Appendix F in Fig. 26.

Backbone cost versus performance. Figure 6 shows, for each backbone architecture, the distribution of mean VUS-PR and mean training time that SPIRAL achieves across the 23 datasets from TSB-AD-U benchmark. Each observation in the violin corresponds to one dataset, so the shape of the distribution reflects how consistently each backbone performs across diverse time series characteristics. PVT-v2-b1-AE achieves the highest mean VUS-PR (0.433), outperforming CNN-AE (0.409) and ResNet18-AE (0.414) with a tighter cross-dataset distribution (std 0.265 vs. 0.283 for CNN-AE), indicating more consistent performance. In terms of training cost, ResNet18-AE is the most efficient backbone for SPIRAL, training 23% faster than CNN-AE (27.8s vs. 36.0s) while achieving comparable VUS-PR. PVT-v2-b1-AE trains 9% faster than CNN-AE (32.8s) while delivering the highest TSAD performance.

It is worth noting that the observed gain of 0.024 in mean VUS-PR of PVT-v2-b1-AE over CNN-AE comes with a 14-fold increase in parameter count (15.39M vs. 1.10M), though not with higher training cost, as shown above. This is because most additional parameters belong to the ImageNet-pretrained encoder, while the decoder/bottleneck trained in all pretrained variants is small and shared, and the frozen/progressive-unfreezing or lower-rate schedules reduce the effective optimization burden relative to training the CNN-AE fully from scratch. Placed in context, the performance gap between backbone architectures (0.024 VUS-PR points between CNN-AE and PVT-v2-b1-AE for SPIRAL) is nearly ten times smaller than the gap across TS2I transformations (0.232 VUS-PR points between SPIRAL and the naive



(a) Mean VUS-PR distribution per backbone for SPIRAL shown as violin plot.

(b) Mean training time (s) distribution per backbone for SPIRAL shown as violin plot.

Fig. 6 Distribution of mean VUS-PR (a) and mean training time in seconds (b) for SPIRAL across backbone architectures ($n = 23$ datasets) shown as violin plots. Each observation corresponds to one evaluation dataset. Horizontal bar refers to the median; diamond refers to the mean. Equivalent plots for all TS2I transformations and metrics are provided in online Appendix F

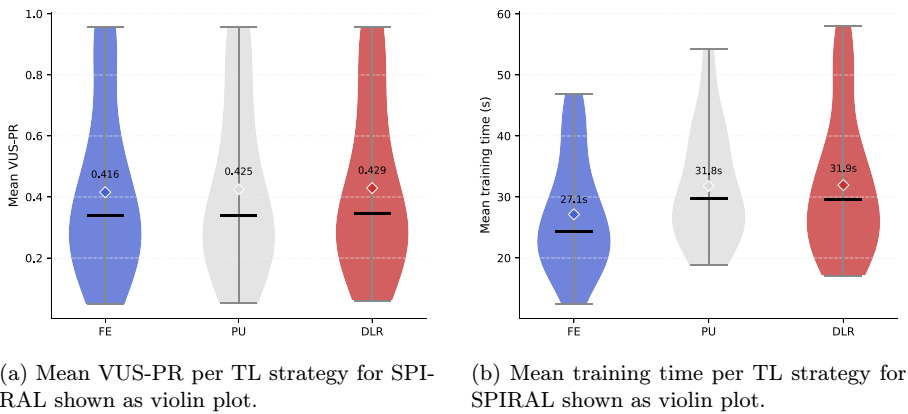


Fig. 7 Distribution of mean VUS-PR (a) and mean training time (b) across transfer learning strategies for SPIRAL ($n = 23$ datasets) shown as violin plots. Each observation corresponds to one evaluation dataset. Horizontal bar refers to median; diamond refers to mean. Equivalent plots for all TS2I transformations and metrics are provided in online Appendix G

RN baseline), confirming that the choice of input representation governs detection performance far more than model capacity. For resource-constrained deployments, ResNet18-AE offers the most favorable efficiency–accuracy trade-off; CNN-AE is a viable alternative at comparable VUS-PR. PVT-v2-b1-AE is appropriate when the priority is maximizing mean detection performance and cross-dataset consistency, and the additional model complexity is acceptable.

These results answer our **RQ4**: training dynamics are primarily governed by the quality of the TS2I transformation, not by convergence speed or backbone capacity. Among the considered TS2I transformations, our SPIRAL method combines the best detection performance with competitive convergence and stable training dynamics, and its superiority is robust to the choice of backbone architecture. Among the evaluated configurations, we observe that ResNet18-AE offers the most favourable trade-off between training efficiency and detection accuracy.

5.4 Transfer Learning Strategies for Vision Backbones

To streamline the discussion, this section consequently focuses on VUS-PR as the primary evaluation metric; analogous figures for all remaining eight metrics and all TS2I transformations are provided in online Appendix G.

Figure 7 shows the distribution of mean VUS-PR and mean training time across transfer learning strategies for SPIRAL, the best-performing TS2I transformation. Differential learning rate (DLR) achieves the highest mean VUS-PR (0.429), marginally outperforming progressive unfreezing (PU, 0.425) and frozen encoder (FE, 0.416). This ordering – $DLR \geq PU \geq FE$ – is consistent across all ten TS2I transformations (see online Appendix G), suggesting that allowing all layers to adapt during fine-tuning is universally beneficial.

Cost versus performance trade-off. The performance advantage of DLR over FE is modest: 1.4 percentage points in VUS-PR for SPIRAL, and only 1.1 points globally across all TS2I transformations. This marginal gain comes at a non-trivial computational cost –

DLR requires 31.9s per dataset for SPIRAL compared to 27.1s for FE, an overhead of 18%. PU offers a similar performance gain to DLR (1.0pp over FE globally) at a comparable cost (31.8s). Crucially, the performance gap across TL strategies (0.011 VUS-PR points globally) is an order of magnitude smaller than the gap across TS2I transformations (0.232 points), reinforcing the data-centric conclusion from Sect. 5.3: the choice of input representation dominates over fine-tuning strategy.

In general, DLR converges slightly slower than FE (31.5 vs. 29.7 epochs) but achieves the highest mean VUS-PR (0.321), while FE is the fastest to converge at the cost of the lowest detection performance (0.310). PU occupies an intermediate position on both axes. The differences are subtle – all three strategies cluster within a 2-epoch and 1.1 percentage point range – confirming that TL strategy has a minor influence on training dynamics relative to the choice of TS2I transformation.

Answering our **RQ5**, DLR yields the highest VUS-PR across all evaluated configurations and is therefore the recommended transfer learning strategy for vision-based TSAD. However, given the small absolute performance difference, FE offers a compelling alternative when training efficiency is a priority – particularly for SPIRAL, where FE is 18% faster than DLR while sacrificing only 1.4 percentage points in VUS-PR. Notably, the consistent positive gain of fine-tuning over frozen encoders is particularly meaningful given the domain gap between the pretraining data and the target domain: ResNet18 and PVT-v2-b1 encoders were pretrained on ImageNet, whose natural images are fundamentally different in structure and statistics from the time-series images produced by TS2I transformations. The fact that even FE achieves competitive performance suggests that low-level visual features learned on ImageNet transfer well to TS2I representations, while the additional gains from PU and DLR confirm that task-specific adaptation of higher-level features further improves TSAD performance.

6 Conclusions

In this work, we introduced a TSAD-oriented workflow with ACF-guided windowing and proposed SPIRAL, a computationally efficient TS2I transformation method tailored to anomaly detection. Across the TSB-AD-U benchmark, SPIRAL consistently delivered the strongest anomaly-detection performance among evaluated TS2I alternatives while remaining among the fastest transformations in preprocessing throughput. The empirical study supports three main conclusions. First, TS2I-based vision models with SPIRAL are not only competitive with strong time-domain baselines, but achieve better cross-dataset consistency in overall ranking. Second, SPIRAL's richer representation does not impose prohibitive computational overhead. Third, among ten TS2I mappings, SPIRAL yields the best average detection quality. The practical implications extend beyond raw TSAD evaluation metrics: SPIRAL achieves the highest VUS-PR while maintaining stable convergence, whereas its nearest competitor suffers from 40% higher training instability – a critical consideration for real-world anomaly detection systems and online/continual learning settings.

This study evaluates SPIRAL exclusively on univariate time series. As noted in Sect. 2, this follows the convention of all ten TS2I transformations benchmarked here: each maps a single scalar-valued window to a 2D image without explicitly modeling cross-variable dependencies (Ni et al., 2025). Extending to multivariate settings is an open challenge for

the entire TS2I paradigm (Li et al., 2023; Oh et al., 2025), and the TS2I transformation step is only one component of this challenge. A complete multivariate workflow must also address channelization, i.e., how to compose K scalar-valued series into a spatial image compatible with pretrained RGB vision backbones, as well as window selection across heterogeneous channels and score aggregation across variables. For TSAD in particular, this challenge extends further: the resulting image representation must not only capture cross-variable structure but also preserve and amplify the subtle, often localized deviations that characterize anomalies – deviations that may span multiple channels or become apparent only through cross-channel relationships. Additionally, all experiments employ reconstruction-based autoencoders at a fixed image resolution of 64×64 , and point-level anomaly scores are obtained by propagating each window's reconstruction error uniformly to its constituent time points, to align with the TSB-AD benchmark evaluation protocol (Liu et al., 2024); this uniform propagation smooths event boundaries and may reduce precision on segment-level metrics such as PA-F1 and Event-based-F1. The sensitivity of results to image resolution and the suitability of alternative vision paradigms such as one-class classification or contrastive learning remain open directions for future investigation.

Open avenues for future work include extending vision-based TSAD to multivariate settings, investigating learnable TS2I representations via contrastive pretraining, evaluating online adaptation under distribution shift and concept drift, and broadening validation across additional domains and streaming settings.

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Data Availability All datasets used in this study are publicly available as part of the TSB-AD benchmark at <https://github.com/TheDatumOrg/TSB-AD>.

Declarations

Conflict of interest The authors declare that they have no competing interests.

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Consent to Participate Not applicable.

Consent for Publication Not applicable.

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